DAY 1 HANDS-ON LABS BRIEF · SESSIONS 014 & 015

TP1 & TP2: Tool Calling and Grounded Document Search

Building a Scope 3 & EU CBAM Carbon Tax Agent in the morning (`Session 014`) and an EU Greenwashing Auditor in the afternoon (`Session 015`) with `gemma-4-26b-a4b-it` & `gemma-4-31b-it` .

INSTRUCTOR	PROGRAM & COHORT	SCHEDULE
Chouaieb Nemri	Year 4 · BSc Data Science	Wednesday, October 14, 2026

Overview of Day 1 Labs (Starter Énoncés on GitHub `main`)

Pull the latest `main` branch (`git pull origin main`). During Day 1, your repository contains the **Starter notebooks (`_Starter_*.ipynb`)** and JSON datasets in `03_Labs_TP/` . We will debrief the solutions live at the end of each block, and full reference solutions will be published via GitHub Release at the end of the course.

Lab 1 (`TP1` , `10:20–11:40`): Scope 3 ESG & EU CBAM Carbon Tax Agent

- **Starter Notebook:** `03_Labs_TP/TP1_ReAct_and_Function_Calling/TP1_Starter_ReAct_and_Tools.ipynb`
- **Dataset:** `03_Labs_TP/data/supplier_esg_audit_db.json` (10 industrial suppliers: `SUP-001` to `SUP-010`)
- **Model:** `MODEL_ID = "gemma-4-26b-a4b-it"` (or `"gemma-4-31b-it"`)

Your 3 Exercises + Stretch Task in TP1

1. **Exercise 1 — Structured Outputs with Pydantic (`SupplierQuickAssessment`):**
   - Complete `SupplierQuickAssessment(BaseModel)` with `supplier_id` , `company_name` , `total_scope1_2_tco2e` , `risk_level` (`Literal["LOW_RISK", "MEDIUM_RISK", "HIGH_RISK"]`), and `key_findings` (`list[str]`).
   - Call `client.models.generate_content` with `response_mime_type="application/json"` and `response_schema=SupplierQuickAssessment` on `SUP-002` (`BalkanFerro Heavy Works`).
2. **Exercise 2 — Design 3 Typed Python Tools (`TOOL_REGISTRY`):**
   - `lookup_supplier_emissions(supplier_id: str) -> dict` : Returns `{"status": "success", "supplier": ...}` or a self-healing error dict listing valid IDs (`SUP-001` .. `SUP-010`) when queried for `SUP-999` .
   - `calculate_cbam_tax_liability(scope1_tco2e, scope2_tco2e, eu_cbam_covered, allocation_factor=0.15, carbon_price_eur=85.0) -> dict` : Computes exact EU CBAM exposure in Euros.
   - `check_csrd_compliance_status(annual_spend_eur, scope1_tco2e, scope2_tco2e, sbti_verified, labor_safety_incidents_12m) -> dict` : Evaluates CSRD Policy v4.2 (`LOW_RISK` , `MEDIUM_RISK` , `HIGH_RISK`).
3. **Exercise 3 — Build the Manual ReAct Loop (`run_manual_react_agent`):**
   - Disable Automatic Function Calling (`AutomaticFunctionCallingConfig(disable=True)`) so you control the loop up to `max_iterations=5` .
   - Use the pre-built `strip_thought_parts(response.candidates[0].content)` helper when appending model turns to `contents` (saves 65% of Free-Tier input tokens!).
   - Dispatch requested tools from `TOOL_REGISTRY` and feed results back via `types.Part.from_function_response(...)` .
4. **Stretch Exercise — Carbon Price Stress-Test (`€85` vs. `€110/ton`):**
   - Implement `compare_carbon_price_scenarios(supplier_id="SUP-002", rates=[85.0, 110.0])` to quantify the extra CBAM tax liability if EU ETS carbon prices spike to `€110/tCO2e` .

Lab 2 (`TP2` , `15:10–17:00`): EU Greenwashing Auditor with Grounded RAG

- **Starter Notebook:** `03_Labs_TP/TP2_Grounded_RAG_and_Interactions_API/TP2_Starter_Grounded_Agentic_RAG.ipynb`
- **Dataset:** `03_Labs_TP/data/green_claims_corpus.json` (8 EU directives & corporate sustainability disclosures)
- **Models:** `gemma-4-26b-a4b-it` (or `gemma-4-31b-it`) + `gemini-embedding-2`

The Problem We Are Solving

Corporations often advertise products as *"100% Carbon Neutral"* (`CLAIM-AERO-101`) or *"Zero-Impact"* (`CLAIM-FASTFASH-103`) based on cheap voluntary offsets or unverified blends. Under **EU Directive 2024/825 (`REG-EU-GCD-01`)**, generic offset-based neutrality claims are prohibited. Your agent must retrieve both the corporate filing and the governing EU regulation, evaluate compliance, and cite exact `doc_id` s.

Your 2 Exercises + Stretch Task in TP2

1. **Exercise 1 — Semantic Retrieval & EU Rule Evaluation Tools:**
   - Implement `search_sustainability_corpus(query: str, category_filter: str = "ALL") -> dict` over `DOCUMENTS` (`Regulation` vs. `Corporate Filing`).
   - Implement `evaluate_eu_green_claim_rule(relies_on_voluntary_offsets, offset_share_pct, has_third_party_iso_or_sbti_audit, scope1_change_pct) -> dict` returning `"VERIFIED_COMPLIANT"` , `"NON_COMPLIANT_GREENWASHING"` , or `"UNSUBSTANTIATED_CLAIM"` .
2. **Exercise 2 — Stateful Interactions API & 2-Stage Grounded Audit (`audit_corporate_claim`):**
   - Test stateful server-side memory using `client.interactions.create(model=MODEL_ID, previous_interaction_id=...)` and inspect `interaction.output_text` .
   - Define `GreenwashingAuditVerdict(BaseModel)` (`company_evaluated` , `headline_claim` , `verdict` , `cited_doc_ids` , `evidence_summary`).
   - Implement `audit_corporate_claim(company_query: str) -> GreenwashingAuditVerdict` using the provided `strip_thought_parts()` and `generate_with_retry()` helpers:
     - *Stage 1:* Run the tool-calling loop with `search_sustainability_corpus` and `evaluate_eu_green_claim_rule` .
     - *Stage 2:* Call `generate_with_retry` with `response_mime_type="application/json"` and `response_schema=GreenwashingAuditVerdict` .
3. **Stretch Exercise — Batch Compliance Table Across Corporate Claims:**
   - Loop over `["AeroTransit carbon neutral", "NordicVolt 72% lower carbon", "VelvetLoop zero-impact", "RhôneAlpin 58% reduction"]` and print a formatted compliance table with verified `cited_doc_ids` .

> **End-of-Day Checkoff (17:15 – 18:15 on Wednesday, Oct 14)**
>
> At `17:00` , we will review the `TP2` execution trace together and launch the **Graded Individual Homework (`04_Homework_Oct14_to_Oct21/`)**. Before disconnecting at `18:15` , every student will run `check_milestone_1()` in `Homework_Starter_Responsible_Supply_Chain_Agent.ipynb` to verify their local setup is 100% ready.